

Paper_Numbers

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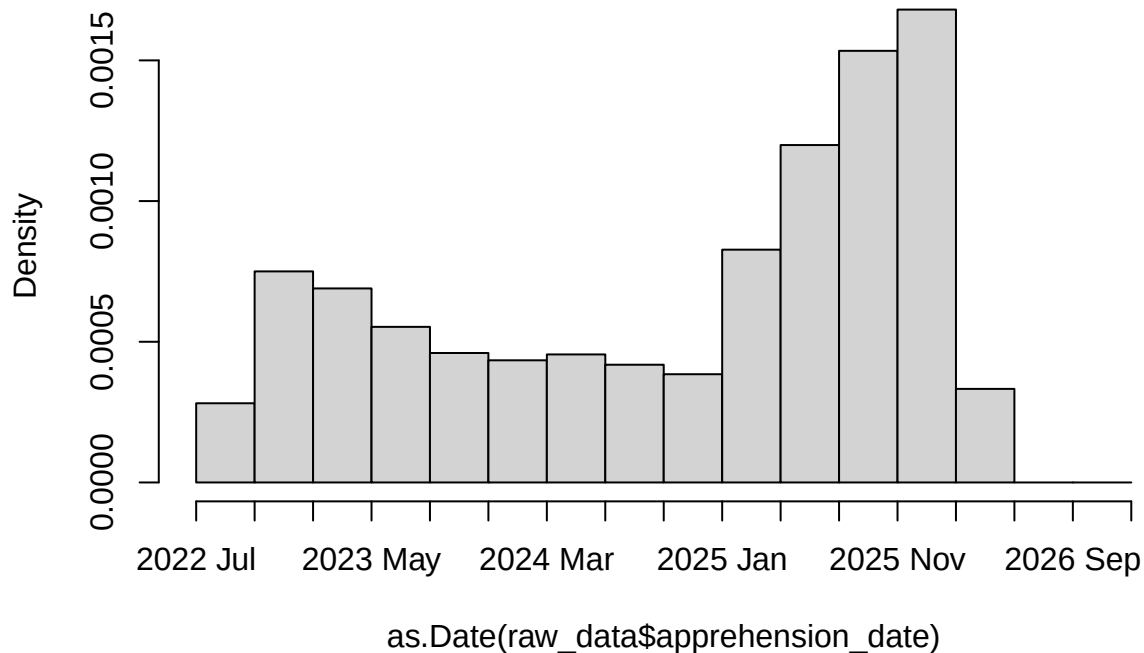
Our primary data source is administrative from the Deportation Data Project (DDP) obtained through FOIA litigation, on the universe of ICE Enforcement and Removal Operations from [DATE to DATE] (n=XXX).

```
raw_data <- read_parquet("../data/raw/arrests-latest.parquet")
```

```
# plot hist of apprehension dates
```

```
hist(as.Date(raw_data$apprehension_date), breaks = 20)
```

Histogram of as.Date(raw_data\$apprehension_date)



```
# sum restricted number of total arrests
```

```
raw_data <- raw_data %>%
```

```
  filter((as.Date(apprehension_date) >= as.Date("2022-10-01")) &  
         (as.Date(apprehension_date) <= as.Date("2026-03-10")))
```

```
nrow(raw_data)
```

```
## [1] 713462
```

Dates run from 10-01-2022 to 03-10-2026 We have a total of 713464 arrests that occur in these dates (2 were outside of them in the raw data)

Restrictions of the data We restrict the sample to exclude minors ($n =$), arrestees with unknown state of arrest ($n =$); see SI.

```
MCA = c('MEXICO', 'GUATEMALA', 'EL SALVADOR', 'HONDURAS', 'NICARAGUA', 'COSTA RICA', 'PANAMA', 'BELIZE')

CARIBBEAN = c('CUBA', 'DOMINICAN REPUBLIC', 'HAITI', 'JAMAICA', 'TRINIDAD AND TOBAGO', 'BAHAMAS', 'BARBADOS',
  'ANTIGUA AND BARBUDA', 'SAINT LUCIA', 'GRENADA', 'DOMINICA', 'SAINT VINCENT AND THE GRENADINES',
  'SAINT KITTS AND NEVIS', 'ST. LUCIA', 'ST. KITTS-NEVIS', 'ST. VINCENT-GRENADINES', 'ANTIGUA-BARBUDA',
  'BRITISH VIRGIN ISLANDS', 'TURKS AND CAICOS ISLANDS', 'BERMUDA', 'NETHERLANDS ANTILLES', 'GUADELOUPE',
  'CURACAO', 'ARUBA', 'ANGUILLA', 'MONTserrat', 'CAYMAN ISLANDS', 'SINT MAARTEN(DUTCH)', 'SINT EUSTATIUS'
)
SA = c('BRAZIL', 'COLOMBIA', 'VENEZUELA', 'ECUADOR', 'PERU', 'ARGENTINA', 'BOLIVIA', 'CHILE', 'PARAGUAY',
  'URUGUAY', 'GUYANA', 'SURINAME', 'FRENCH GUIANA')
ASIA = c(
  'CHINA', 'INDIA', 'PHILIPPINES', 'VIETNAM', 'SOUTH KOREA', 'JAPAN', 'BANGLADESH', 'PAKISTAN',
  'INDONESIA', 'MALAYSIA', 'THAILAND', 'SRI LANKA', 'NEPAL', 'BURMA', 'MYANMAR', 'CAMBODIA', 'LAOS',
  'SINGAPORE', 'MONGOLIA', 'TAIWAN', 'AFGHANISTAN', 'UZBEKISTAN', 'KAZAKHSTAN', 'KYRGYZSTAN', 'TAJIKISTAN',
  'TURKMENISTAN', 'NORTH KOREA', 'BHUTAN', 'MALDIVES', 'BRUNEI', 'IRAN', 'IRAQ', 'SYRIA', 'LEBANON', 'JORDAN',
  'ISRAEL', 'YEMEN', 'SAUDI ARABIA', 'TURKEY', 'ARMENIA', 'AZERBAIJAN', 'GEORGIA', 'KUWAIT', 'QATAR',
  'BAHRAIN', 'UNITED ARAB EMIRATES', 'OMAN', 'PALESTINE',
  # DDP-specific spellings:
  'CHINA, PEOPLES REPUBLIC OF', 'TURKIYE', 'KOREA', 'HONG KONG', 'MACAU', 'EAST TIMOR',
  'PALESTINE BORN BEFORE 1948'
)
AFRICA = c(
  'NIGERIA', 'ETHIOPIA', 'EGYPT', 'KENYA', 'GHANA', 'SOMALIA', 'SUDAN', 'SOUTH SUDAN', 'SOUTH AFRICA',
  'SENEGAL', 'CAMEROON', 'LIBERIA', 'SIERRA LEONE', 'ERITREA', 'IVORY COAST', 'COTE D'IVOIRE', 'MALI',
  'MOROCCO', 'TUNISIA', 'ALGERIA', 'UGANDA', 'TANZANIA', 'ZIMBABWE', 'RWANDA', 'BURKINA FASO', 'CHAD',
  'NIGER', 'GAMBIA', 'GUINEA', 'TOGO', 'BENIN', 'MAURITANIA', 'ANGOLA', 'MOZAMBIQUE', 'ZAMBIA', 'MALAWI',
  'MADAGASCAR', 'BOTSWANA', 'NAMIBIA', 'LESOTHO', 'CONGO', 'DEMOCRATIC REPUBLIC OF THE CONGO',
  'CENTRAL AFRICAN REPUBLIC', 'EQUATORIAL GUINEA', 'GABON', 'COMOROS', 'CAPE VERDE', 'DJIBOUTI',
  'BURUNDI', 'LIBYA',
  # DDP-specific spellings:
  'DEM REP OF THE CONGO', 'GUINEA-BISSAU', 'MAURITIUS', 'SAO TOME AND PRINCIPE', 'ESWATINI'
)
EUCA = c(
  'CANADA', 'UNITED KINGDOM', 'IRELAND', 'GERMANY', 'FRANCE', 'ITALY', 'SPAIN', 'PORTUGAL', 'GREECE',
  'RUSSIA', 'UKRAINE', 'POLAND', 'ROMANIA', 'HUNGARY', 'CZECH REPUBLIC', 'SLOVAKIA', 'BULGARIA', 'SERBIA',
  'CROATIA', 'SLOVENIA', 'MACEDONIA', 'NORTH MACEDONIA', 'ALBANIA', 'BOSNIA AND HERZEGOVINA', 'BELGIUM',
  'NETHERLANDS', 'SWEDEN', 'NORWAY', 'DENMARK', 'FINLAND', 'ICELAND', 'SWITZERLAND', 'AUSTRIA',
  'LUXEMBOURG', 'CYPRUS', 'MALTA', 'ESTONIA', 'LATVIA', 'LITHUANIA', 'BELARUS', 'MOLDOVA', 'MONTENEGRO',
  'KOSOVO', 'AUSTRALIA', 'NEW ZEALAND', 'FIJI', 'SAMOA', 'TONGA', 'PAPUA NEW GUINEA',
  # DDP-specific spellings:
  'BOSNIA-HERZEGOVINA', 'USSR', 'YUGOSLAVIA', 'SERBIA AND MONTENEGRO', 'CZECHOSLOVAKIA', 'ANDORRA',
  'MONACO', 'MICRONESIA, FEDERATED STATES OF', 'MARSHALL ISLANDS', 'PALAU', 'FRENCH POLYNESIA'
)

get_region <- function(country){
  if (country %in% MCA){
    return("MCA")
  } else if (country %in% CARIBBEAN){
    return("CARIBBEAN")
  } else if (country %in% SA){
```

```

    return("SA")
  } else if (country %in% ASIA){
    return("ASIA")
  } else if (country %in% AFRICA){
    return("AFRICA")
  } else if (country %in% EUCA){
    return("EUCA")
  } else{
    return(NA)
  }
}

get_method <- function(method){
  if (method %in% LEA_METHODS){
    return("LEA")
  } else if (method %in% CA_METHODS){
    return("CA")
  } else{
    return("OTHER")
  }
}

LEA_METHODS = c('287(g) Program','Anti-Smuggling','CAP Federal Incarceration','CAP Local Incarceration',
  'CAP State Incarceration','Criminal Alien Program','ERO Reprocessed Arrest',
  'Law Enforcement Agency Response Unit','Organized Crime Drug Enforcement Task Force',
  'Other Agency (turned over to INS)','Other Task Force','Probation and Parole',
  'Custodial Arrest','Patrol Border','Patrol Interior')

CA_METHODS = c('Located','Non-Custodial Arrest','Worksite Enforcement')

STATES = c(toupper(state.name), "DISTRICT OF COLUMBIA")

raw_arrests <- read_parquet("../data/raw/arrests-latest.parquet") %>%
  rename("ApprehensionDate" = "apprehension_date",
    "ApprehensionAOR" = "apprehension_aor",
    "ApprehensionMethod" = "apprehension_method",
    "ApprehensionCriminality" = "apprehension_criminality"
  ) %>%
  select("ApprehensionDate", "ApprehensionAOR",
    "ApprehensionMethod", "ApprehensionCriminality", "apprehension_state",
    "citizenship_country", "gender", "birth_year", "case_threat_level")

raw_arrests <- raw_arrests %>%
  mutate(aor_short = sub(" Area of Responsibility", "", ApprehensionAOR))

raw_arrests[is.na(raw_arrests$apprehension_state),]$apprehension_state = ""

raw_arrests$state_clean = raw_arrests$apprehension_state

STATES = c(toupper(state.name), "DISTRICT OF COLUMBIA")
raw_arrests[!(raw_arrests$state_clean %in% STATES),]$state_clean = ""

raw_arrests <- raw_arrests |>

```

```

mutate(state_clean = case_when(
  aor_short == "Phoenix" ~ "ARIZONA",
  aor_short %in% c("Buffalo", "New York City") ~ "NEW YORK",
  aor_short %in% c("San Diego", "Los Angeles") ~ "CALIFORNIA",
  aor_short %in% c("Houston", "Harlingen", "San Antonio", "Dallas") ~ "TEXAS",
  aor_short == "Miami" ~ "FLORIDA",
  TRUE ~ state_clean
))

raw_arrests[is.na(raw_arrests$aor_short),]$aor_short = ""
raw_arrests[raw_arrests$aor_short == "HQ",]$aor_short = ""

raw_arrests <- raw_arrests |>
  mutate(aor_short = case_when(
    aor_short == "" & state_clean %in% c("ALASKA", "WASHINGTON", "OREGON") ~ "Seattle",
    aor_short == "" & state_clean %in% c("HAWAII") ~ "San Francisco",
    aor_short == "" & state_clean %in% c("UTAH", "NEVADA", "IDAHO", "MONTANA") ~ "Salt Lake City",
    aor_short == "" & state_clean %in% c("ARIZONA") ~ "Phoenix",
    aor_short == "" & state_clean %in% c("NEW MEXICO") ~ "El Paso",
    aor_short == "" & state_clean %in% c("OKLAHOMA") ~ "Dallas",
    aor_short == "" & state_clean %in% c("WYOMING", "COLORADO") ~ "Denver",
    aor_short == "" & state_clean %in% c("UTAH", "NEVADA", "IDAHO", "MONTANA") ~ "Salt Lake City",
    aor_short == "" & state_clean %in% c("MINNESOTA", "IOWA", "NEBRASKA", "SOUTH DAKOTA", "NORTH DAKOTA", "KANSAS", "MISSOURI", "ILLINOIS", "INDIANA", "KENTUCKY", "MISSISSIPPI", "ALABAMA", "ARKANSAS", "LOUISIANA", "TENNESSEE", "MISSISSIPPI", "ALABAMA", "FLORIDA") ~ "Miami",
    aor_short == "" & state_clean %in% c("GEORGIA", "SOUTH CAROLINA", "NORTH CAROLINA") ~ "Atlanta",
    aor_short == "" & state_clean %in% c("VIRGINIA", "DISTRICT OF COLUMBIA") ~ "Washington",
    aor_short == "" & state_clean %in% c("MICHIGAN", "OHIO") ~ "Detroit",
    aor_short == "" & state_clean %in% c("MARYLAND") ~ "Baltimore",
    aor_short == "" & state_clean %in% c("NEW JERSEY") ~ "Newark",
    aor_short == "" & state_clean %in% c("DELAWARE", "PENNSYLVANIA", "WEST VIRGINIA") ~ "Philadelphia",
    aor_short == "" & state_clean %in% c("VERMONT", "NEW HAMPSHIRE", "MAINE", "MASSACHUSETTS", "CONNECTICUT") ~ "Boston",
    TRUE ~ aor_short
  ))

# obtain count of minors:
print("Minors:")

## [1] "Minors:"

sum(((year(ymd(raw_arrests$ApprehensionDate)) - raw_arrests$birth_year) < 18), na.rm = T)/nrow(raw_arrests)

## [1] 4.193765

arrests <- raw_arrests %>%
  filter((as.Date(ApprehensionDate) >= as.Date("2023-11-27")) &
    (as.Date(ApprehensionDate) <= as.Date("2026-03-10"))) %>%
  filter((year(ymd(ApprehensionDate)) - birth_year) >= 18) %>% # filter out children
  select(c(ApprehensionDate, ApprehensionMethod,
    ApprehensionCriminality, case_threat_level,
    state_clean, aor_short, citizenship_country)) %>%
  mutate(citizenship_region = unlist(lapply(citizenship_country, get_region)))

# missing location information

```

```

print("Missing location information")

## [1] "Missing location information"
sum((raw_arrests$state_clean == "") | (raw_arrests$aor_short == ""))/nrow(raw_arrests) *100

## [1] 7.674949
arrests <- arrests %>%
  filter(state_clean != "" & aor_short != "")

print("Region of Origin Unknown:")

## [1] "Region of Origin Unknown:"
sum(is.na(arrests$citizenship_region))/nrow(raw_arrests) *100

## [1] 0.04092708
arrests <- arrests %>%
  filter(!is.na(citizenship_region))

```

We test each of these claims using data on more than half a million administrative ICE arrests from [DATE] to March 10, 2026 (the most recent date currently available) from the Deportation Data Project (CITE). To do so, we first temporally aggregate arrests into 14 60-day periods, anchored at January 20, 2025. Our study period thus ranges from TKTK to TKTK pre-treatment, and from TKTK to TKTK post-treatment (n = TKTK).

```

INAUG_DATE = "2025-01-20"
WINDOW_DAYS = 60
TOTAL_DAYS_EACH_SIDE = 7*WINDOW_DAYS

assign_window <- function(dates){
  days_from_inaug = as.numeric(as.Date(dates) - as.Date(INAUG_DATE))
  window_assignment = ifelse(days_from_inaug < 0, -((-days_from_inaug - 1) %/% WINDOW_DAYS + 1),
                             days_from_inaug %/% WINDOW_DAYS)
  window_assignment = ifelse(abs(days_from_inaug) > TOTAL_DAYS_EACH_SIDE, NA, window_assignment)
  return(window_assignment)
}

arrests$window <- assign_window(arrests$ApprehensionDate)

print("Start of first window:")

## [1] "Start of first window:"
as.Date("2025-01-20")-60*7

## [1] "2023-11-27"
print("End of final window:")

## [1] "End of final window:"
as.Date("2025-01-20")+60*7-1

## [1] "2026-03-15"
print("Max date available in data:")

## [1] "Max date available in data:"

```

```

max((arrests %>% filter(window == max(arrests$window, na.rm = T)) )$ApprehensionDate)

## [1] "2026-03-10"
print("Outside of window:")

## [1] "Outside of window:"
nrow(arrests %>% filter(is.na(window)))

## [1] 0
nas <- arrests %>%
  filter(is.na(window))
print("Inside of window:")

## [1] "Inside of window:"
arrests <- arrests %>%
  filter(!is.na(window))
nrow(arrests)

## [1] 488362
summary(as.factor(do.call(rbind, lapply(arrests$ApprehensionMethod, get_method))))

##      CA      LEA    OTHER
## 210607 270795   6960
# load raw arrests data to get the other variables
print(paste0("Number of arrests: ", nrow(raw_arrests)))

## [1] "Number of arrests: 713464"
raw_arrests <- raw_arrests %>%
  filter((as.Date(ApprehensionDate) >= as.Date("2023-11-27")) &
    (as.Date(ApprehensionDate) <= as.Date("2026-03-10")))
print(paste0("Filtering to dates of interest: ", nrow(raw_arrests)))

## [1] "Filtering to dates of interest: 522996"
raw_arrests <- raw_arrests %>%
  filter(((year(ymd(ApprehensionDate)) - birth_year) >= 18))
print(paste0("Remove minors: ", nrow(raw_arrests)))

## [1] "Remove minors: 512086"
raw_arrests <- raw_arrests %>%
  filter(state_clean != "")
print(paste0("Remove no state: ", nrow(raw_arrests)))

## [1] "Remove no state: 488984"
raw_arrests <- raw_arrests %>%
  filter(aor_short != "")
print(paste0("Remove no aor: ", nrow(raw_arrests)))

## [1] "Remove no aor: 488654"
raw_arrests_no_region <- raw_arrests %>%
  mutate(citizenship_region = unlist(lapply(citizenship_country, get_region))) %>%

```

```

filter(is.na(citizenship_region))

raw_arrests <- raw_arrests %>%
  mutate(citizenship_region = unlist(lapply(citizenship_country, get_region))) %>%
  filter(!is.na(citizenship_region))

print(paste0("Require valid region: ", nrow(raw_arrests)))

```

```
## [1] "Require valid region: 488362"
```

Figure out how many AORs mismatch their states

```

valid_state_aor <- raw_arrests %>%
  rename("aor" = "aor_short",
         "state" = "state_clean") %>%
  select(aor, state)

valid_state_aor <- valid_state_aor %>%
  filter((aor == "Seattle" & state %in% c("ALASKA", "WASHINGTON", "OREGON")) |
        (aor == "San Francisco" & state %in% c("CALIFORNIA", "HAWAII")) |
        (aor == "San Diego" & state == "CALIFORNIA") |
        (aor == "Los Angeles" & state == "CALIFORNIA") |
        (aor == "Salt Lake City" & state %in% c("UTAH", "NEVADA", "IDAHO", "MONTANA")) |
        (aor == "Phoenix" & state == "ARIZONA") |
        (aor == "El Paso" & state %in% c("TEXAS", "NEW MEXICO")) |
        (aor == "San Antonio" & state == "TEXAS") |
        (aor == "Harlingen" & state == "TEXAS") |
        (aor == "Houston" & state == "TEXAS") |
        (aor == "Dallas" & state %in% c("TEXAS", "OKLAHOMA")) |
        (aor == "Denver" & state %in% c("WYOMING", "COLORADO")) |
        (aor == "St. Paul" & state %in% c("MINNESOTA", "IOWA", "NEBRASKA", "SOUTH DAKOTA", "NORTH DAKOTA")) |
        (aor == "Chicago" & state %in% c("WISCONSIN", "ILLINOIS", "INDIANA", "KENTUCKY", "MISSOURI", "ARIZONA")) |
        (aor == "New Orleans" & state %in% c("ARKANSAS", "LOUISIANA", "TENNESSEE", "MISSISSIPPI", "ALABAMA")) |
        (aor == "Miami" & state == "FLORIDA") |
        (aor == "Atlanta" & state %in% c("GEORGIA", "SOUTH CAROLINA", "NORTH CAROLINA")) |
        (aor == "Washington" & state %in% c("VIRGINIA", "DISTRICT OF COLUMBIA")) |
        (aor == "Detroit" & state %in% c("MICHIGAN", "OHIO")) |
        (aor == "Baltimore" & state == "MARYLAND") |
        (aor == "Newark" & state == "NEW JERSEY") |
        (aor == "Philadelphia" & state %in% c("DELAWARE", "PENNSYLVANIA", "WEST VIRGINIA")) |
        (aor == "New York City" & state == "NEW YORK") |
        (aor == "Buffalo" & state == "NEW YORK") |
        (aor == "Boston" & state %in% c("VERMONT", "NEW HAMPSHIRE", "MAINE", "MASSACHUSETTS", "CONNECTICUT")))

invalid_state_aor <- raw_arrests %>%
  rename("aor" = "aor_short",
         "state" = "state_clean")

invalid_state_aor <- invalid_state_aor %>%
  filter(!((aor == "Seattle" & state %in% c("ALASKA", "WASHINGTON", "OREGON")) |
        (aor == "San Francisco" & state %in% c("CALIFORNIA", "HAWAII")) |
        (aor == "San Diego" & state == "CALIFORNIA") |
        (aor == "Los Angeles" & state == "CALIFORNIA") |
        (aor == "Salt Lake City" & state %in% c("UTAH", "NEVADA", "IDAHO", "MONTANA")) |

```

```

(aor == "Phoenix" & state == "ARIZONA") |
(aor == "El Paso" & state %in% c("TEXAS", "NEW MEXICO")) |
(aor == "San Antonio" & state == "TEXAS") |
(aor == "Harlingen" & state == "TEXAS") |
(aor == "Houston" & state == "TEXAS") |
(aor == "Dallas" & state %in% c("TEXAS", "OKLAHOMA")) |
(aor == "Denver" & state %in% c("WYOMING", "COLORADO")) |
(aor == "St. Paul" & state %in% c("MINNESOTA", "IOWA", "NEBRASKA", "SOUTH DAKOTA", "NORTH DAKOTA")) |
(aor == "Chicago" & state %in% c("WISCONSIN", "ILLINOIS", "INDIANA", "KENTUCKY", "MISSOURI", "ARIZONA")) |
(aor == "New Orleans" & state %in% c("ARKANSAS", "LOUISIANA", "TENNESSEE", "MISSISSIPPI", "ALABAMA")) |
(aor == "Miami" & state == "FLORIDA") |
(aor == "Atlanta" & state %in% c("GEORGIA", "SOUTH CAROLINA", "NORTH CAROLINA")) |
(aor == "Washington" & state %in% c("VIRGINIA", "DISTRICT OF COLUMBIA")) |
(aor == "Detroit" & state %in% c("MICHIGAN", "OHIO")) |
(aor == "Baltimore" & state == "MARYLAND") |
(aor == "Newark" & state == "NEW JERSEY") |
(aor == "Philadelphia" & state %in% c("DELAWARE", "PENNSYLVANIA", "WEST VIRGINIA")) |
(aor == "New York City" & state == "NEW YORK") |
(aor == "Buffalo" & state == "NEW YORK") |
(aor == "Boston" & state %in% c("VERMONT", "NEW HAMPSHIRE", "MAINE", "MASSACHUSETTS", "CONNECTICUT"))

```

```

invalid_state_aor_most_common <- invalid_state_aor %>%
  group_by(aor, state) %>%
  summarize(n = n()) %>%
  arrange(-n)

```

```

## `summarise()` has regrouped the output.
## i Summaries were computed grouped by aor and state.
## i Output is grouped by aor.
## i Use `summarise(groups = "drop_last")` to silence this message.
## i Use `summarise(by = c(aor, state))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.

```